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Exercise 1

A. Describe the map followed by a reduce parallel pattern providing PRAM complexity and show at least one example discussed during the lectures. (1)

B. Please provide a practical example of a loop carry dependency that cannot be parallelizable. (1)

C. Given the following C-based code:

```
int sum = 0;

for (i = 4; i < 100; i++) {
   for (j = 1; j < 100; j++) {
      x[i][j] = g(x[i-4][j], y[i][j-1]);
   }
}

for (i = 0; i < 100; i++) {
   for (j = 1; j < 100; j++) {
      z[i][j-1] = z[i][j-1] + x[i][j-1] * w[i][j-1];
      sum *= z[i][j-1];
   }
}
```

where matrices x, y, z, and w store integers and already contain valid data before the execution of the code.
Discuss when function g prevents code parallelization (a practical example is required). Is the memory layout of x,y,z creating issues for the parallelization of the code (show an example where this is going to happen)? How and under what conditions can the previous code be parallelized? (2)

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Exercise 2

A. Please describe how the Unpack parallel pattern works and its performance considering a PRAM model. How is it implemented (pseudo code required)? (2)

B. A program processes a large array in two phases. In the first phase, the same independent operation is applied to every element. In the second phase, all processed values are combined into a single scalar result. How would you structure this computation in OpenMP, and what correctness or performance problems could arise from an implementation using atomic primitives? (2)

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Exercise 3

For all the points in this exercise, consider the following C snippet of code that prints all the elements in a tree (the order of lines in the output does not matter):

```
typedef struct node {
 int value;
 struct node *left;
 struct node *right;
} node;

void dfs(node *root) {
 if (root == NULL)
   return;

 printf("%d\n", root->value);

 dfs(root->left);
 dfs(root->right);
}

Int main() {
  node                                                                    root;
  // initialize the tree (code omitted)
  dfs(&root);
}
```

A. Why can't we use the OpenMP sections construct to parallelize the code? (1)

B. Which is a suitable OpenMP worksharing construct to use for parallelizing the code? (1)

C. Describe exactly which OpenMP pragma you would use to make the code parallel. You can either decorate the code or write here what pragma you use and where to put them (2)

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Exercise 4

A. Make an application example where the usage of the tag parameter in MPI_Recv matters. (1) **This question can be skipped if you got 1 or 3 points in the second challenge.**

B. In an application that uses only MPI to parallelize the computation, it would make sense to use a mutex? Explain the reason. (1)

C. How can I use OpenMP, or multithreading in general, to work synergistically with MPI? If we decide to use only MPI, what features of the standard should I use to overlap computation with communication within a single MPI Process? Please explain also the reasons (2) **This question can be skipped if you got 2 or 3 points in the second challenge.**